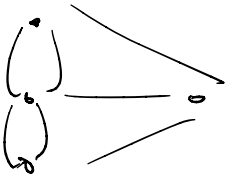
Last time:

Definition. *Let* G *be a multigraph.*

- *An* ***Eulerian circuit*** *is a circuit in* G *that traverses each edge exactly once and uses all vertices.*
- *An* ***Eulerian trail*** *is an open trail in* G *that traverses each edge exactly once and uses all vertices.*

Lemma. *Let* G *be a multigraph, and let*

$$T = (v_0, e_1, v_1, e_2, \dots, v_{k-1}, e_k, v_k)$$



be a trail in G*. Fix* v *any vertex in* T*.*

(1) If T *is a circuit (i.e.* $v_0 = v_k$*), then* T *contains an even number of edges incident to* v*.*

(2) If T *is open and* $v \neq v_0, v_k$*, then* T *contains an even number of edges incident to* v*.*

(3) If T *is open and* $v = v_0, v_k$*, then* T *contains an odd number of edges incident to* v*.*

Theorem (6.1)**.** *Let* G *be a multigraph. Then* G *is Eulerian if and only if* G *is connected and all vertices have even degree.*

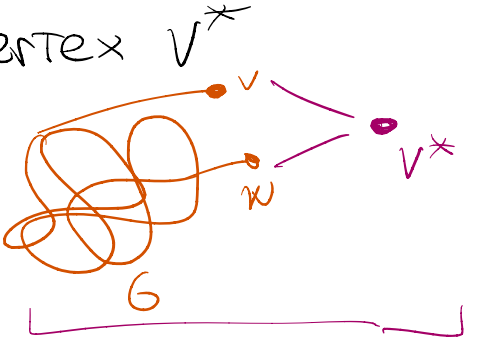
Theorem (6.2). *Let* G *be a multigraph. Then* G *has an Eulerian trail if and only if* G *is connected and has exactly two odd-degree vertices.*

Proof: $\Rightarrow$) Exercise: Follow $(\Rightarrow)$ in 6.1, but use parts (2), (3) of lemma

$\Leftarrow$) Let v, w the vertices in G with odd degree.

Let G' obtained from G by adding new vertex v^* and edges v^*v, v^*w. Then

1) G' is connected
2) All vertices in G' have even degree.



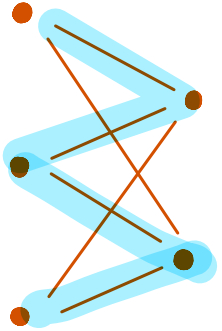
By Thm 6.1, G' contains an Eulerian circuit C starting at v^*.

But then $C - v^*v - v^*w$ is an Eulerian Trail in G.

Definition. *Let* G *be a graph.*

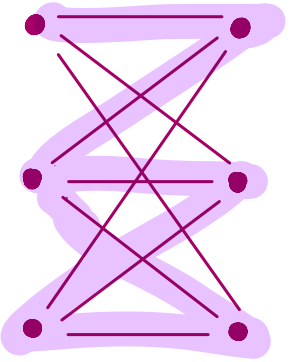
- *A* ***Hamiltonian cycle*** *is a cycle in* G *that uses all vertices.* H-cycle
- *A* ***Hamiltonian path*** *is a path in* G *that uses all vertices.* H-path
- G *is a* ***Hamiltonian graph*** *if it contains a Hamiltonian cycle.*

Example. *Decide if the following graphs contain a Hamiltonian cycle or a Hamiltonian path.* P= Petersen

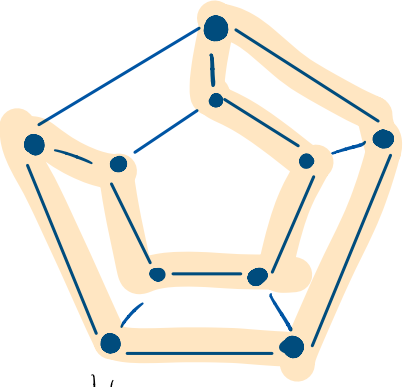


H-path

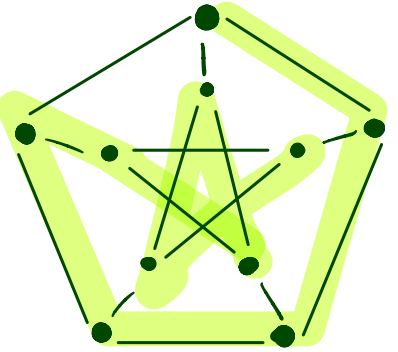
not H-cycle



H-cycle



H-cycle



H-path

H-cycle?

Some first remarks: Assume G is a graph on $n \geqslant 3$ vertices and C is a Hamiltonian cycle in G. Then

(a) Deleting any edge of C gives a Hamiltonian path in G.

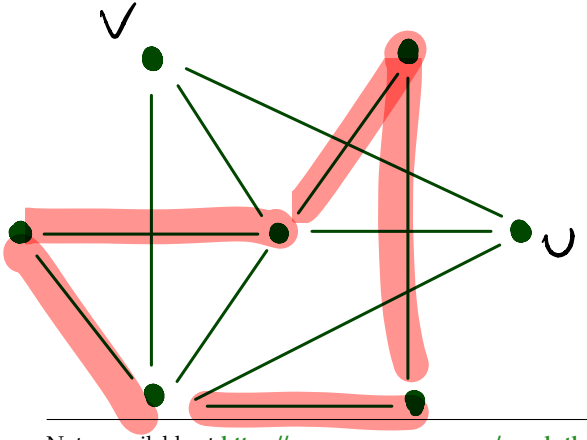
(b) $C \cong C_n$ is a cycle of length n. Thus G is obtained from C_n by adding some additional edges.

(c) $C \cong C_n$ contains no cycles of length less than n

(d) Each $v \in V(G)$ has exactly two neighbors in C.

(e) If $v \in V(G)$ has $\deg_G v = 2$, then both edges incident to v lie on C.

Example. *The following graph is non-Hamiltonian:*

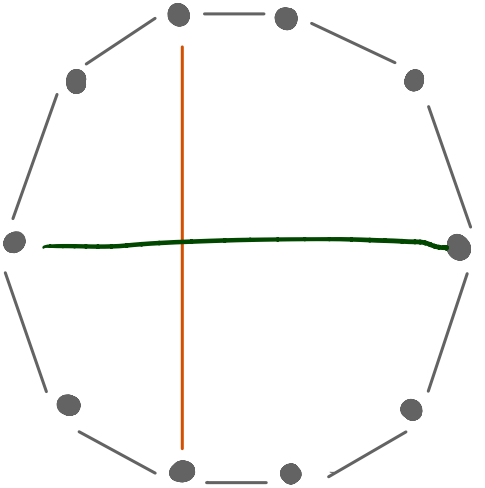


Come from e)

But now can't incorporate u, v.

Example. *The Petersen graph is non-Hamiltonian:*

C_{10}



- P is 3-regular on 10 vertices

Thus if P is Hamiltonian, we can get from C_{10} by 5 edges.

- There are no cycles of length 3 nor 4 in P.

Can only add 1 edge

So P non Hamiltonian

Can add 2 edges

Next: Unlike Eulerianity, deciding if a graph is Hamiltonian is *hard*. (NP complete)
We give several sufficient conditions for Hamiltonianity.

Theorem (6.5). *If G is a Hamiltonian graph, then for every nonempty proper set $U \subsetneq V(G)$ we have*

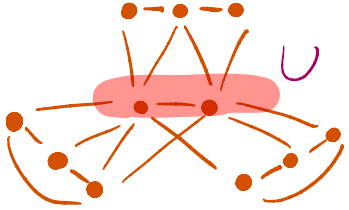
$$\text{Number of components of } G - U \leq |U|$$

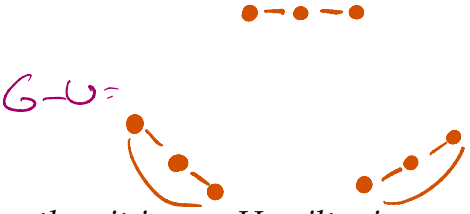
Proof: Let $C = (v_0, v_1, \ldots, v_n = v_0)$ H-cycle in G

Let $U \subseteq V(G)$, $U \neq \emptyset$, $U \neq V(G)$. Let $G_1, \ldots, G_k$ denote the components of $G-U$. Fix some G_i. Let j be the last index such that $v_j \in G_i$. Then v_{j+1} → not in G_i ($j+1 > j$)
→ not in other G_h no! no edges between G_i and G_h. Thus $v_{j+1} \in U$. Thus for each component G_i we have an element in U

Thus $k \leq |U|$

Example. *The following graph is non-Hamiltonian*





Corollary. *If G contains a cut vertex, then it is non-Hamiltonian.*

Exercise. *Write down the contrapositive to Theorem 6.5. Prove the Corollary*

Theorem (6.6, Ore's condition). *Let* G *be a graph on* $n \geq 3$ *vertices. If*

$$\deg u + \deg v \geq n$$

for each pair u, v *of nonadjacent vertices, then* G *is Hamiltonian.*

Proof: Fix some n. Assume Thm not true for this n. So among all counterexamples, take one G, with maximum number of edges

Thus (1) $\deg u + \deg v \geq n$ whenever $uv \notin E(G)$

(2) G non-Hamiltonian.

(3) If $xy \notin E(G)$ Then $G + xy$ is Hamiltonian.

Note: G is NOT complete. (For $n \geq 3$, K_n is Hamiltonian: $V(K_n) = \{v_1, \ldots, v_n\}$ Then $(v_1, v_2, \ldots, v_n, v_1)$ is H-cycle)

Thus, There are some $x, y \in V(G)$ such that $xy \notin E(G)$.

By (3) $G+xy$ is Hamiltonian contains a Hamiltonian cycle C which must use xy (G non-Hamiltonian)

Now $P = C - xy$ is an H-path in G, with endpoints x, y.

$$P = (x = x_1, x_2, \dots, x_n = y)$$

Let $S = \{x_i : 2 \le i \le n : x x_i \in E(G)\}$, $|S| = \deg x$

$T = \{x_i : 2 \le i \le n : x_{i-1} y \in E(G)\}$ $|T| = \deg y$

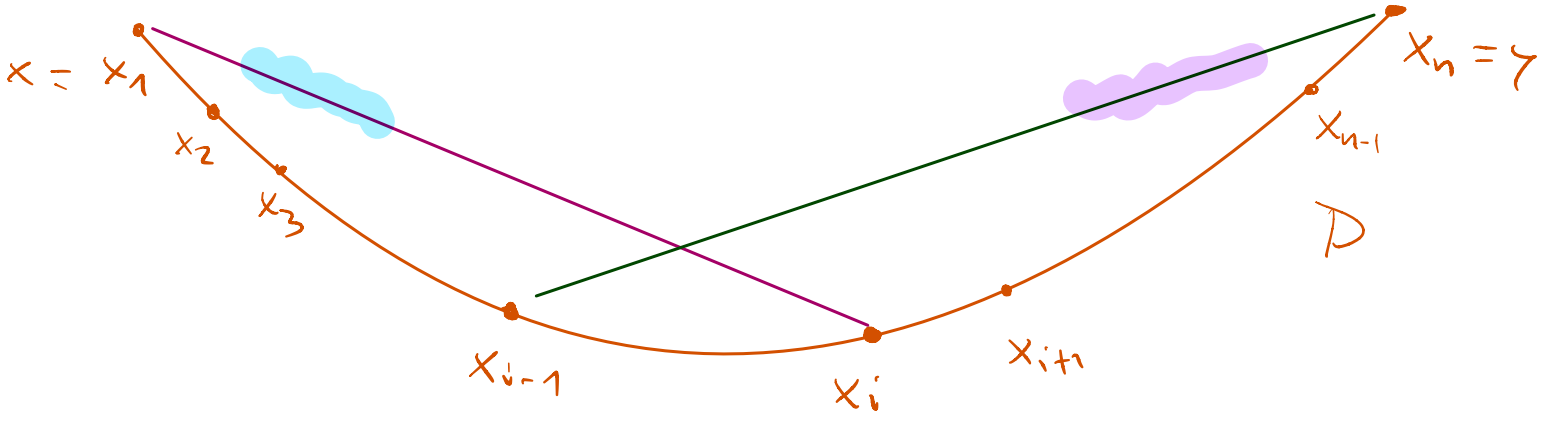
And $S \cup T \subseteq \{x_2, \dots, x_n\}$, Thus $|S \cup T| \le n-1$

Claim: $S \cap T \neq \emptyset$

$$|S \cap T| = |S| + |T| - |S \cup T| = \deg x + \deg y - |S \cup T|$$
$$\ge n - |S \cup T| \ge n - (n-1) = 1$$

Hence there is some $2 \le i \le n$ such that

$x x_i \in E(G)$ and $x_{i-1} y \in E(G)$



BUT THEN

Portion of P

Portion of P (backwards)

$(x_1, x_i, x_{i+1}, x_{i+2}, \ldots, x_{n-1}, x_n, x_{i-1}, x_{i-2}, \ldots, x_2, x_1)$

is a Hamiltonian cycle in G, contradiction.



Corollary (6.7, Dirac's condition). *Let* G *be a graph on* $n \geqslant 3$ *vertices. If* $\deg v \geqslant n/2$ *for each vertex* v*, then* G *is Hamiltonian.*

Proof: $u, v \in V(G)$ Then

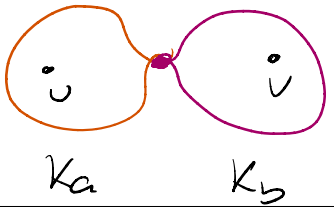
$\deg u + \deg v \geqslant \frac{n}{2} + \frac{n}{2} = n$ apply Ore's condition

Exercise. *If* $n \geqslant 3$*, then the clique* K_n *is Hamiltonian.*

Exercise. *Let* $s, t \geqslant 2$*. The complete bipartite graph* $K_{s,t}$ *is Hamiltonian if and only if* $s = t$.

Exercise (The bound in Ore's condition is sharp). *For any* n*, show that there is a non-Hamiltonian graph* G *on* n*-vertices such that* $\deg u + \deg v \geqslant n - 1$ *for each pair* u, v *of nonadjacent vertices.*

$n \geq 4$



$a + b - 1 = n$

$\deg u = a - 1$

$\deg v = b - 1$

$a + b - 2 = n - 1$

Theorem (6.8, Bondy-Chvátal condition). *Let G be a graph on $n \geqslant 3$ vertices. Assume u, v are nonadjacent vertices such that $\deg u + \deg v \geqslant n$. Then G is Hamiltonian if and only if $G + uv$ is Hamiltonian.*

Proof: $\Rightarrow$) Clear: an H-cycle in G is an H-cycle in G+uv

$\Leftarrow$) Assume G+uv is Hamiltonian. For a contradiction

Assume G is non Hamiltonian

Thus in G+uv we have an H-cycle that uses uv

Then in G we have H-path with endpoints u,v.

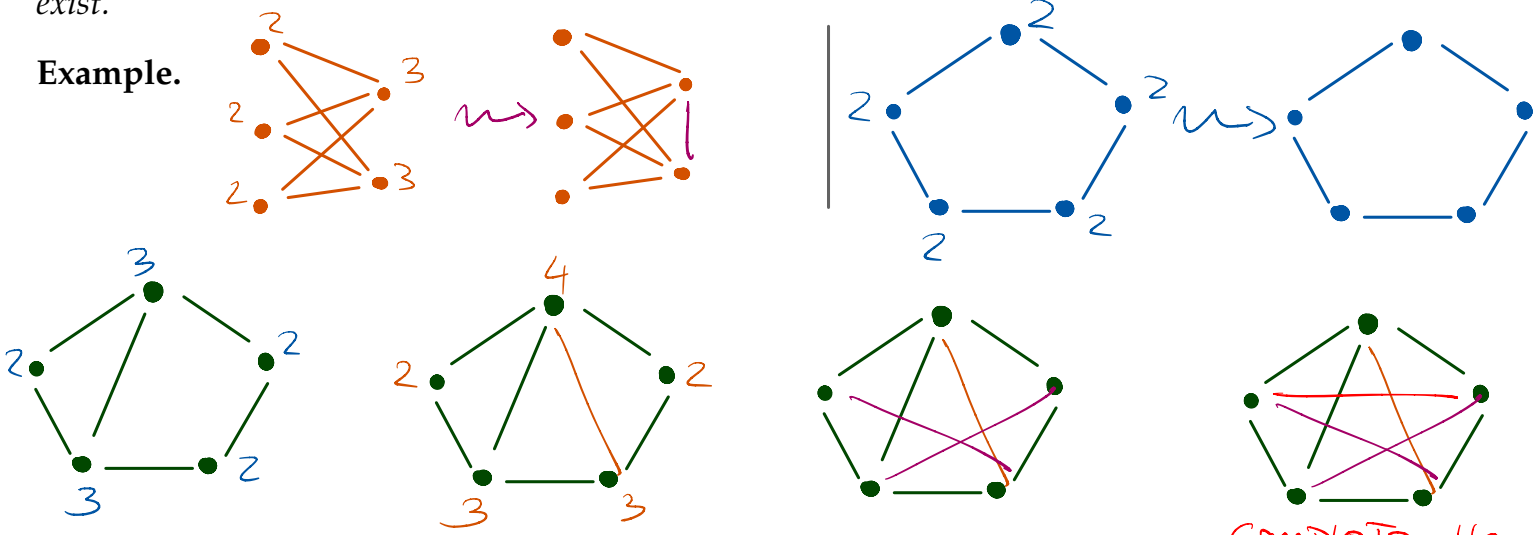
As in the proof of Ore's condition, using $\deg_G u + \deg_G v \geq n$,

we can build an H-cycle in G, contradiction

(G non Hamiltonian) ∎

Definition. *Let G be a graph on n vertices. The closure $C(G)$ of G is the graph obtained from G by recursively adding edges uv whenever u, v are nonadjacent vertices such that $\deg u + \deg v \geqslant n$, until no such vertices exist.*

Example.



Corollary (6.9). *A graph is Hamiltonian if and only if its closure is Hamiltonian.*

Proof: Repeated applications of 6.8. ∎

For the last theorem, let us denote

$$N_j(G) = |\{v \in V(G) : \deg v \leqslant j\}| = \text{number of vertices with degree at most } j$$

Theorem (6.11, Pósa's condition). *Let* G *be a graph on* $n \geqslant 3$ *vertices.*
If for each integer j *with* $1 \leqslant j < n/2$ *we have*

$$N_j(G) < j$$

then G *is Hamiltonian.*

"Proof": Show $C(G)$ is complete.